# Quiz 6

Graded

**Student**

施明睿

**Total Points**

37 / 52 pts

**Question 1**

True or False **24** / 30 pts

1.1 a **3** / 3 pts

✔ – **0 pts** F

– **3 pts** incorrect

1.2 b **3** / 3 pts

✔ – **0 pts** T

– **3 pts** incorrect

1.3 c **3** / 3 pts

✔ – **0 pts** T

– **3 pts** incorrect

1.4 d **3** / 3 pts

✔ – **0 pts** F

– **3 pts** incorrect

1.5 e **3** / 3 pts

✔ – **0 pts** T

– **3 pts** incorrect

1.6 f **3** / 3 pts

✔ – **0 pts** T

– **3 pts** incorrect

1.7 g **3** / 3 pts

✔ – **0 pts** T

– **3 pts** incorrect

1.8 h **0** / 3 pts

– **0 pts** F

✔ – **3 pts** incorrect

1.9 i **3** / 3 pts

✔ – **0 pts** F

– **3 pts** incorrect

1.10 j **0** / 3 pts

– **0 pts** T

✔ – **3 pts** incorrect

Question 2

## Find parent node

9 / 10 pts

- ✔ **+ 5 pts** $P(1)$ correct
- **+ 4 pts** $P(1)$ partially correct
- **+ 3 pts** $P(1)$ partially correct
- **+ 2 pts** $P(1)$ partially correct
- **+ 1 pt** $P(1)$ partially correct
- **+ 5 pts** $P(2)$ correct
- ✔ **+ 4 pts** $P(2)$ partially correct
- **+ 3 pts** $P(2)$ partially correct
- **+ 2 pts** $P(2)$ partially correct
- **+ 1 pt** $P(2)$ partially correct
- **+ 0 pts** Too many incorrect

Question 3

## Lets code

4 / 12 pts

- **+ 6 pts** (a) correct
- **+ 4 pts** (a) partially correct
- ✔ **+ 2 pts** (a) partially correct
- **+ 6 pts** (b) correct
- **+ 4 pts** (b) partially correct
- ✔ **+ 2 pts** (b) partially correct
- **+ 0 pts** Incorrect

CS 101A Fall 2025 - Quiz 6
December 30, 2025 - 20 Minutes

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**1. (30 points) True or False**

(a) (3') On a connected, directed graph with only positive edge weights, Bellman-Ford runs asymptotically as fast as Dijkstra. ○ True ● False

(b) (3') Suppose we want to find the shortest path between all pairs of vertices on graph $G = (V, E)$, where $|E| = 3|V|$ and all edge weights are positive. In this case, Dijkstra's may runs faster than Floyd-Warshall algorithm. ● True ○ False

(c) (3') Given a directed graph $G = (V, E)$, where $V = \{v_1, \ldots, v_n\}$, and $G$ has no negative cycle . In Bellman-Ford's algorithm, after $k$ out-most iterations, the shortest path from $v_1$ to $v_n$ that consists of at most $k$ edges is computed. ● True ○ False

(d) (3') After applying Bellman-Ford's algorithm on node $v$, if there are no negative cycles, we have the minimum distance between any two different nodes $v_i$ and $v_j$. ○ True ● False

(e) (3') On a graph with $n$ vertices and $m$ edges, if all edges have positive weights, Bellman-Ford's algorithm uses $O(mn)$ iterations to find the shortest distance path of a single source. ● True ○ False

(f) (3') In the Floyd–Warshall algorithm, if after the $j$th iteration ($j = |V| - 2$) of the outer loop there exists an index $i$ such that $d[i][i] < 0$, then the graph must contain a negative-weight cycle. ● True ○ False

(g) (3') In any connected graph without a negative cycle, A* tree-search algorithm with consistent heuristics can always find the shortest path between two nodes. ● True ○ False

(h) (3') The time complexity of Dijkstra using binary heap is $O(|E|+|V|\ln|V|)$. ● True ○ False

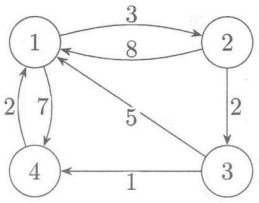
(i) (3') A* Graph Search algorithm returns the optimal shortest path if the heuristic function is admissible. ○ True ● False

(j) (3') In A* graph search algorithm with a consistent heuristic function, if vertex $u$ is marked visited before $v$, then $d(u) + h(u) \le d(v) + h(v)$, where $d(u)$ is the distance from the start vertex to $u$. ○ True ● False

**2. (10 points) Find parent node**

Consider the directed weighted graph $G = (V, E)$ shown below, where $V = \{1, 2, 3, 4\}$ and the vertex order is fixed as $1, 2, 3, 4$. We run the **Floyd–Warshall algorithm** to compute all-pairs shortest paths, and we maintain the matrix $P^{(k)}$, defined as follows: $p_{i,j} = \begin{cases} j, & \text{If there is an edge from } i \text{ to } j, \\ \varnothing, & \text{Otherwise.} \end{cases}$

(a) (10') Write down the matrices $P^{(1)}$ and $P^{(2)}$.




$P^{(0)}$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | ∅ | 8 | 5 | 2 |
| 2 | 3 | ∅ | ∅ | ∅ |
| 3 | ∅ | 2 | ∅ | 2 |
| 4 | 7 | ∅ | 1 | ∅ |

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | ∅ | 1 | 1 | 1 |
| 2 | 2 | ∅ | ∅ | ∅ |
| 3 | ∅ | 3 | ∅ | ∅ |
| 4 | 4 | ∅ | 4 | ∅ |

$P^{(1)}$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | ∅ | 1 | 1 | 1 |
| 2 | 2 | ∅ | 1 | 1 |
| 3 | ∅ | 3 | ∅ | 1 |
| 4 | 4 | 1 | 4 | ∅ |

$P^{(2)}$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | ∅ | 1 | 1 | 1 |
| 2 | 2 | ∅ | 1 | 1 |
| 3 | 2 | 3 | ∅ | 1 |
| 4 | 2 | 1 | 4 | ∅ |

**3. (12 points) Lets code!**

We want to find a single source min distance with Dijkstra's Algorithm and A* **graph search** algorithm. **Suppose all edges have positive weight and the heuristic function is consistent**. The graph mentioned in this problem is a simple directed graph.

**Note:** 'w' is a weight map, where you can get any edge $(u, v)$'s weight by using 'w(u, v)'. 'h' is a consistent heuristic function, you can get the heuristic value of a node $u$ by using 'h(u)'.

dist[i] represents the shortest distance from $s$ to $i$, pre[i] represents the previous node on the shortest path from $s$ to $i$.

$Q$ is a min-heap storing a tuple: (key, value), and sorted by value. And you have the following operations for $Q$:

1. **Q.push({u, val})**: put a tuple (u, val) into the heap.
2. **{u, val} = Q.pop()**: get the tuple with minimum value in the heap, and then pop the tuple out of the heap.
3. **Q.update({u, val})**: find the tuple in the heap whose key is 'u', and update its value into 'val'.

**Algorithm 1** Single Source Shortest Path Algorithm

```
Input: Weight map w, min-heap Q, Source node s, heuristic function h .
Output: The shortest distance from s to all other nodes, and their previous node in the shortest path.
for i ← 0 to V do
    dist[i] ← Inf
    pre[i] ← NULL
    Q.push({i, dist[i]})
end for
dist[s] ← 0
Q.update({s, 0})
while Q is not empty do
    Fill this part with your pseudo code
    ...
end while
return dist, pre
```

What you need to do is to write some **pseudo code** to fill in to implement the algorithms with the given operations.

(a) (6') Implement Dijkstra's algorithm.

(b) (6') Implement A* **graph search** algorithm. You can use $h(v)$ to get the heuristic value for any node $v$.

**Hint:** The algorithm should not differ too much from Dijkstra's algorithm.

(a) 取Q.pop的所有邻接出度节点 ui, ~~{ui, min{Q.pop().val + w(Q.pop().key, ui)}}~~

{ui, min{Q.pop().val + w(Q.pop().key, ui), w(s, ui)}}

最后Q.pop().val为最大.

(b) 取Q.pop的所有邻接出度节点 ui, {ui, min{Q.pop().val + w(Q.pop().key, ui), w(s, ui) + h(ui)}}

最后可得